Questions by topic: Properties of pollutants & environmental factors - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows some properties of three pollutants.

Table 1

Pollutant	State at 20 °C	Solubility in water	Solubility in lipids	Environmental half-life
DDT (insecticide)	Solid	Very low	High	About 15 years in soil
Nitrate ions	Dissolved ion	Very high	Very low	Not applicable (not degraded)
Hydrogen cyanide	Gas (denser than air)	High	Low	Hours

0 1 . 1 Define the term **environmental half-life**.

[1 mark]

.....

.....

0 1 . 2 Use Table 1 to explain why DDT is more likely than nitrate ions to biomagnify in a food chain.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 1 . 3 Suggest why the release of hydrogen cyanide at Bhopal, India, in 1984 caused deaths very close to the source rather than over a wide area.

[2 marks]

.....

.....

.....

.....

Do not write outside the box

0 1 . 4 Explain how the high water solubility of nitrate ions affects the area over which they cause pollution and the concentration of the pollution.

[2 marks]

.....

.....

.....

.....

0 1 . 5 Give **two** examples of pollutants that are forms of energy rather than substances.

[2 marks]

1 .....

.....

2 .....

.....

10

0 2 . 1 Explain the difference between a **mutagen**, a **carcinogen** and a **teratogen**.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 2 . 2 Explain why a birth abnormality caused by a teratogen cannot be inherited by future generations, whereas one caused by a mutagen may be.

[2 marks]

.....

.....

.....

.....

Do not write outside the box

0 2 . 3 Distinguish between **somatic** and **gonadic** effects of a mutagen.

[2 marks]

.....

.....

.....

.....

0 2 . 4 Explain what is meant by **synergism** and give an example.

[3 marks]

.....

.....

.....

.....

.....

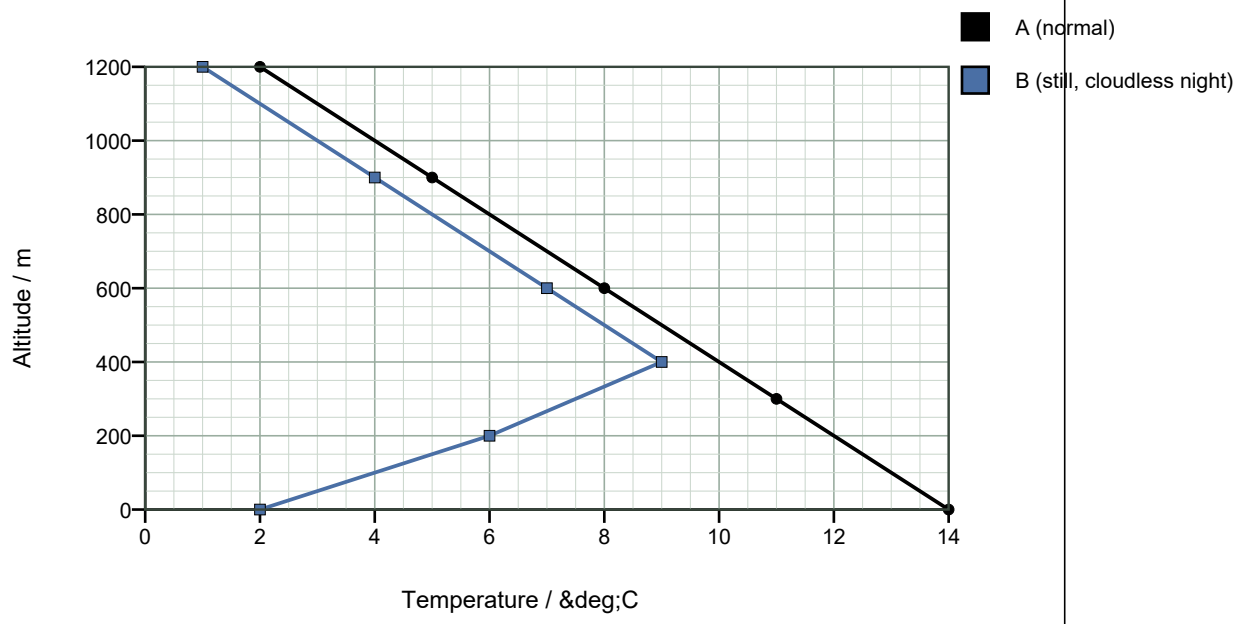
.....

.....

10

0 3 Figure 1 shows how air temperature changes with altitude above a city in a valley on two different nights. Line A shows normal conditions. Line B shows the conditions on a still, cloudless night.

Figure 1



Do not write outside the box

0 3 . 1 Name the atmospheric condition shown by line B in Figure 1.

[1 mark]

.....

.....

0 3 . 2 Use Figure 1 to explain why pollutants released at ground level would disperse on night A but not on night B.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 3 . 3 Explain why the conditions on night B (still, cloudless) allowed an inversion to form.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 3 . 4 Give **two** types of pollution event that are made more severe by temperature inversions.

[2 marks]

Do not write outside the box

1 .....

.....

2 .....

.....

10

0 4 . 1 Explain how each of the following environmental factors affects the rate of degradation of pollutants. Give a named example of a pollutant for each factor.

[6 marks]

Temperature .....

.....

.....

.....

Light .....

.....

.....

.....

Oxygen .....

.....

.....

.....

0 4 . 2 Explain how the presence of adsorbent materials such as clay can reduce the severity of pollution by heavy metals or radioactive caesium-137.

[2 marks]

.....

.....

.....

.....

0 4 . 3 Explain why a fall in pH increases the harm caused by many heavy metals.

[2 marks]

Do not write outside the box

Do not write outside the box

.....

.....

.....

.....

10

0 5 A factory discharges a persistent, water-soluble pollutant into a river. Table 1 shows the concentration of the pollutant measured at sites along the river on a day with normal river flow.

Table 1

Site	Distance downstream of discharge / km	Pollutant concentration / mg dm^{-3}	River flow / m^3 s^{-1}
A (upstream)	-2	0.0	12
B	0.5	4.8	12
C	5	3.9	15
D (below tributary)	12	1.6	36
E	30	1.4	38

0 5 . 1 Calculate the percentage decrease in pollutant concentration between site B and site E.

[2 marks]

Answer ..... %

0 5 . 2 Explain the large decrease in concentration between sites C and D.

[2 marks]

.....

.....

.....

.....

0 5 . 3 Explain why the concentration changes very little between sites D and E.

[2 marks]

Do not write outside the box

.....

.....

.....

.....

0 5 . 4 Suggest why the pollution might be more serious if the same discharge entered a lake instead of a river.

[2 marks]

.....

.....

.....

.....

0 5 . 5 Explain why the factory might be permitted to discharge the effluent into the sea near a strong current but not into the river.

[2 marks]

.....

.....

.....

.....

10

0 6 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 6 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

.....

.....

.....

Do not write outside the box

5

0 7 . 1 Discuss how an understanding of the properties of pollutants can be used to predict their behaviour in the environment and to plan strategies to control them.

[9 marks]

Do not write outside the box

Extra space

0 7 . 2 State what is meant by **bioaccumulation**.

[1 mark]

10

0 8 Mercury released into a lake from a factory was in the inorganic form. Fish in the lake were later found to contain high concentrations of methyl mercury, an organic form.

0 8 . 1 Suggest how the inorganic mercury was converted to methyl mercury.

[2 marks]

.....

.....

.....

.....

0 8 . 2 Explain why methyl mercury is a more hazardous pollutant than inorganic mercury.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 8 . 3 Explain why the health risk from eating fish from the lake may remain for many years after the discharges stop.

[2 marks]

.....

.....

.....

.....

0 8 . 4 Suggest **three** factors that should be included in a Critical Pathway Analysis of a proposed mercury discharge.

[3 marks]

Do not write outside the box

1 .....

.....

2 .....

.....

3 .....

.....

10

END OF QUESTIONS

Do not write outside the box